

Tier	Function / Component	Manufacturer	Manufacturer Part Number	Description / Specification	Quantity	Unit Cost	Total Cost	Interface
2	Aircraft-bus current sensor IC	Infineon	TLE4973A120T5S0001XUMA1	Integrated isolated current sensor; approximately 120/132 A class; automotive-qualified	1			Integrated current path; analog output
2	Bus-voltage measurement	TBD	TBD	Precision high-voltage resistor divider and protection network feeding MCU ADC	1			Analog voltage to ADC
2	Weight-on-Wheels / gear switch	Honeywell	V7-1C13D8-201	SPDT roller-lever microswitch	1	9.62	9.62	Discrete switch input
2	WoW / discrete signal conditioning	Texas Instruments	SN74LVC1G17MDBVREP	Single Schmitt-trigger buffer; Enhanced Product	1	2.5	2.5	Digital GPIO
2	Electrical-monitoring RTOS MCU	Texas Instruments	S5LS20216ASPGEMEP	TMS570LS20216-EP Hercules MCU	1	55.49	55.49	ADC; GPIO; SPI; UART
2	ARINC 429 transmitter	Holt Integrated Circuits	HI-35870PQTF	Single ARINC 429 transmitter IC	1			SPI to MCU; ARINC 429 output
3	Current-sensor output filtering	TBD	TBD	RC anti-alias / noise-filter network between TLE4973 and MCU ADC	1			Analog
3	Bus-input transient protection	TBD	TBD	TVS, series impedance, and clamp network for bus-voltage measurement	1			Power / analog protection
3	Local power regulation	TBD	TBD	Required MCU, sensor, and ARINC voltage rails	1			Power
3	Aircraft external connector	TBD	TBD	Rugged connector for bus current path, bus-voltage sense, WoW, ARINC, and power	1			Aircraft I/O